INCH-POUND
MIL-DTL-24643/59B
27 November 2023
SUPERSEDING
MIL-DTL-24643/59A
28 January 2010

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +176 DEGREES FAHRENHEIT (+80 DEGREES CELSIUS), 150 VOLTS, TYPE LSC5FS (INCLUDING VARIATIONS LSC5FSW, LSC5OS, AND LSC5OSW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in four variations: LSC5FS (metallic foil shield, non-watertight), LSC5FSW (metallic foil shield, watertight), LSC5OS (metallic foil and braid shield, non-watertight), and LSC5OSW (metallic foil and braid shield, watertight).

Construction (Watertight and Non-Watertight)

First - Copper conductor, solid, uncoated, ASTMB3, 24 AWG, 0.0201 inch nominal. Diameter of conductor may vary from nominal as necessary in order for the completed cable to meet the specified electrical requirements.

Second - Polyolefin insulation. Wall thickness as necessary, in order for the completed cable to comply with the electrical requirements specified herein. Maximum insulation diameter shall not exceed 0.046 inch.

Third - Two insulated conductors, twisted together with an appropriate lay length for each pair such that the completed cable complies with the electrical requirements specified herein. Wire identification codes shall be in compliance with ANSI/TIA 568.2 as follows:

Pair 1 – white/blue and blue
Pair 2 – white/orange and orange
Pair 3 – white/green and green
Pair 4 – white/brown and brown

Fourth - Four pairs shall be cabled together with a lay as required to comply with the electrical requirements specified herein. Cabling sequence shall be consecutive. Fillers or other means may be used as necessary to form a firm, well-rounded cable that complies with the electrical requirements specified herein.

Fifth - An optional binder.

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Construction for Variant Types LSC5FS and LSC5FSW

Sixth - Copper drain wire, tin-coated, 24 AWG, solid or stranded, ASTM B286.

Seventh - Metallic foil/polyester tape (metallic foil side facing in and in contact with the drain wire). Tape may be longitudinally or helically applied, with a minimum overlap of 10 percent.

Eighth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.040 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Construction for Variant Types LSC5OS and LSC5OSW

Sixth - Metallic foil/polyester tape (metallic foil side facing out and in contact with the braid). Tape may be longitudinally or helically applied, with a minimum overlap of 10 percent.

Seventh - Braided shield of 36 AWG or 34 AWG, tin-coated copper.

Eighth - An optional binder.

Ninth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.040 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. Details.

Military part no. M24643/59	Type and size	Overall diameter (inch)	
		min.	max.
-01UO	LSC5FS-4	0.240	0.360
-02UO	LSC5FSW-4	0.240	0.380
-03UO	LSC5OS-4	0.290	0.400
-04UO	LSC5OSW-4	0.300	0.420

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/100 meters at 68 °F [20 °C], max.)	9.38
Conductor resistance unbalance (percent, max.)	5
Jacket flaws	No failure
Conductor and shield continuity	No failure
Voltage withstand	
Conductor to conductor	1.5 kVAC or 2.5 kVDC
Conductor to overall shield	1.5 kVAC or 2.5 kVDC

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Insulation resistance (megohms/100 meters, max.)	Not applicable
Group A:	
Visual and dimensional	No failure
Watertightness (types LSC5FSW and LSC5OSW only) (see MIL-DTL-24643 for limits of water leakage)	No failure
Capacitance	
Mutual capacitance (nominal at 1 kilohertz [kHz]) (nanofarad [nF]/100 meters)	5.6
Pair-to-ground capacitance unbalance (max. at 1 kHz) (picofarad [pF]/100 meters)	330
Input impedance (at 1 megahertz [MHz] to 100 MHz) (ohms ±15%)	100
Swept electronic parameters	See table II

TABLE II. Swept electrical parameters.

Frequency (MHz)	1.0	4.0	10	16	31.25	62.5	100
Return loss dB/100 m (min.)	20.0	23.0	25.0	25.0	23.6	21.5	20.1
Insertion loss dB/100 m (max.)	2.0	4.1	6.5	8.2	11.7	17.0	22.0
NEXT dB/100 m (min.)	65.3	56.3	50.3	47.2	42.9	38.4	35.3
ELFEXT dB/100 m (min.)	63.8	51.8	43.8	39.7	33.9	27.9	23.8
PS NEXT dB/100 m (min.)	62.3	53.3	47.3	44.2	39.9	35.4	32.3
PS ELFEXT dB/100 m (min.)	60.8	48.8	40.8	36.7	30.9	24.9	20.8
Propagation delay ns/100 m (max.)	570	552	545	543	540	539	538
Delay skew ns/100 m (max.)	45	45	45	45	45	45	45

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Group B:

Cold bending (cable)	No failure
Drip (203±1.8 °F [95±1 °C]) (types LSC5FSW and LSC5OSW only)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Thermoset proof test (percent, max.)	
Insulation	Not applicable
Jacket (when tested at 392 °F [200 °C])	50

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No failure
Cable fill (sealant) removability (types LSC5FSW and LSC5OSW only)	No failure
Heat distortion (percent of unaged, max.)	30
Permanence of printing (jacket) (cycles, min.)	125
Shield (conformance to material, construction, and coverage) (types LSC5FS and LSC5FSW only)	No failure
Surface transfer impedance (types LSC5OS and LSC5OSW only)	
Milliohms per meter (max.)	700

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Electromagnetic pulse (EMP) response (dB, min.)	60

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (203±3.6 °F [95±2 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	Not applicable
Stress resistance	No failure
Temperature humidity (types LSC5FSW and LSC5OSW only)	No failure

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UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
All types and sizes	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-075)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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